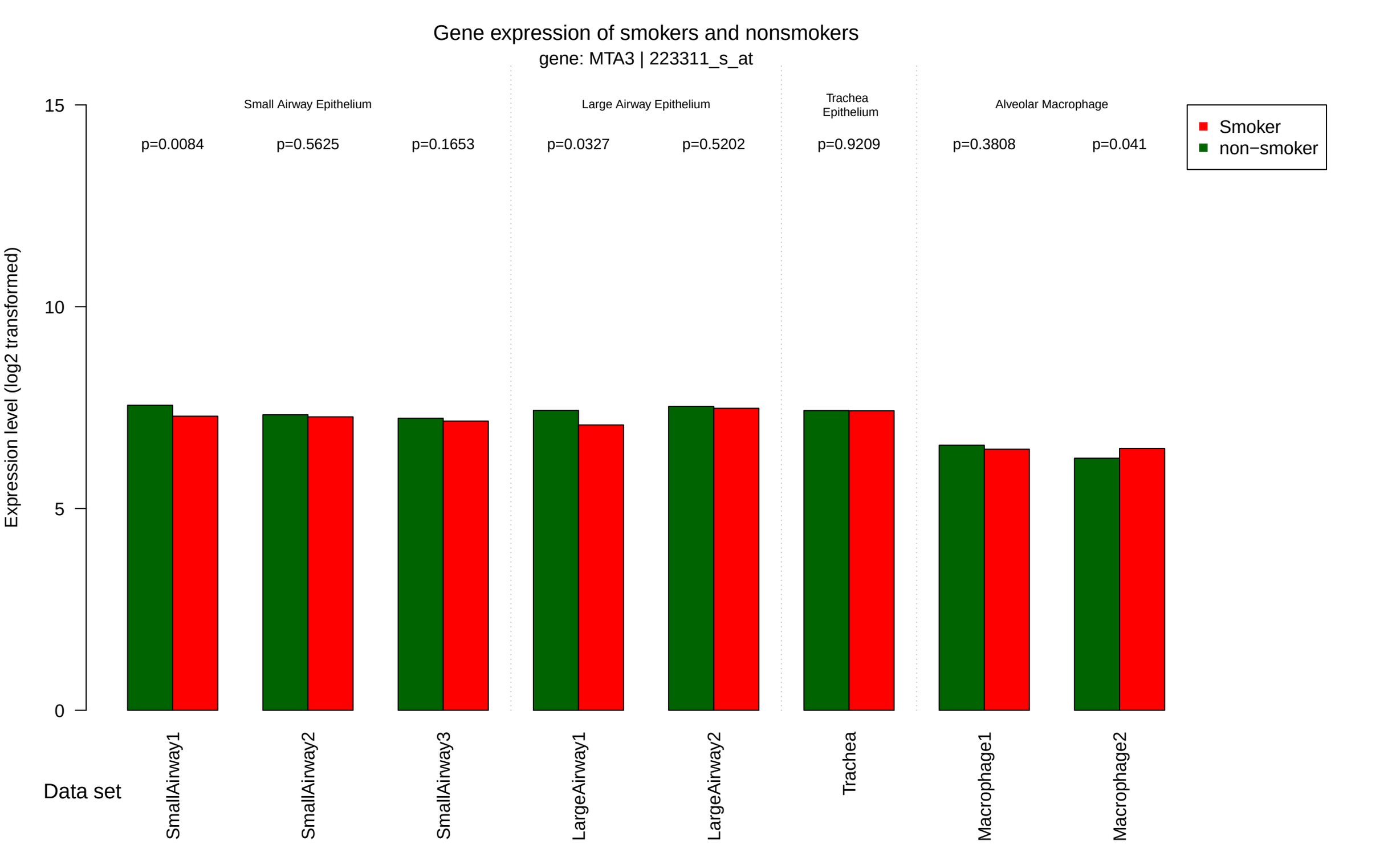
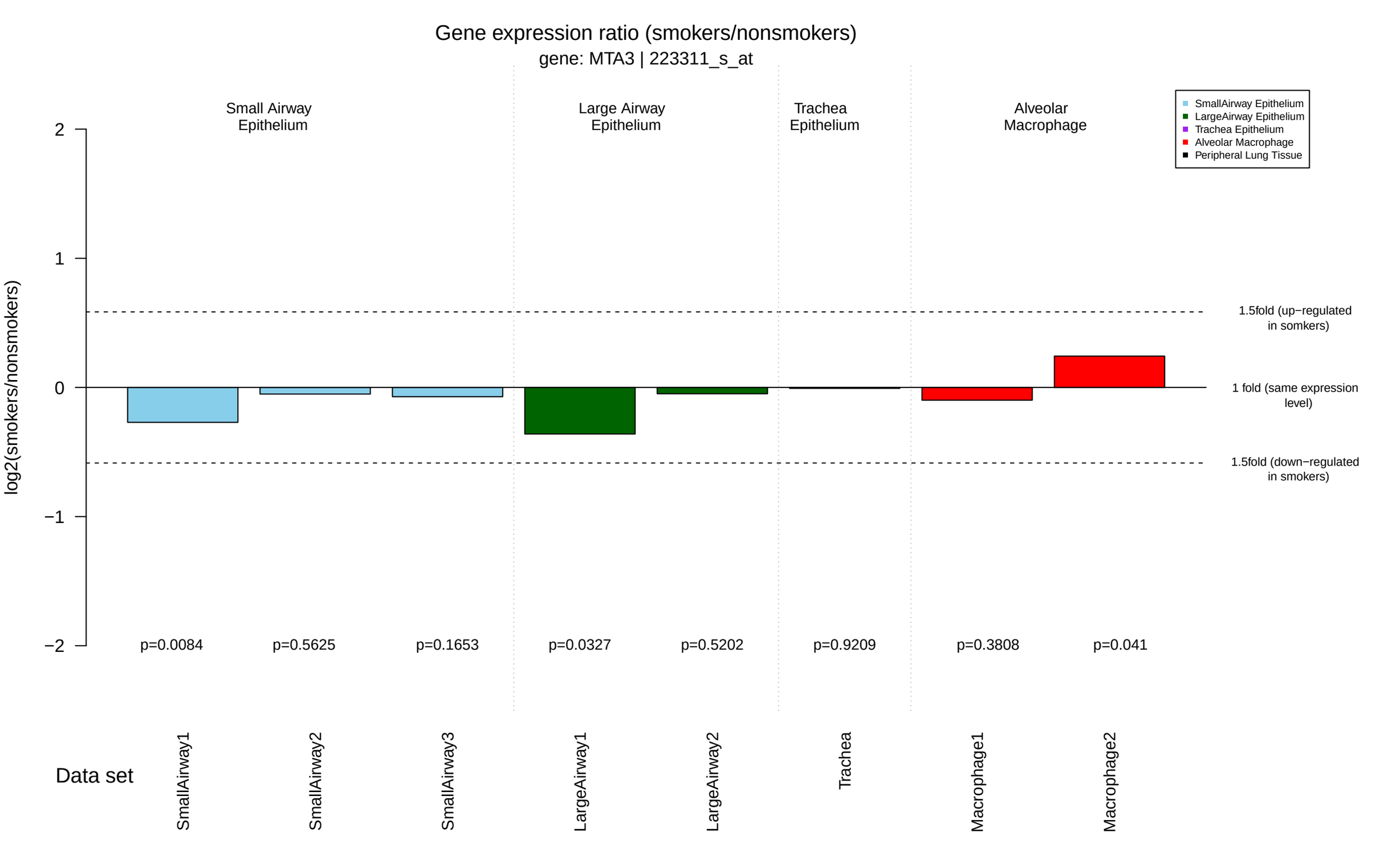
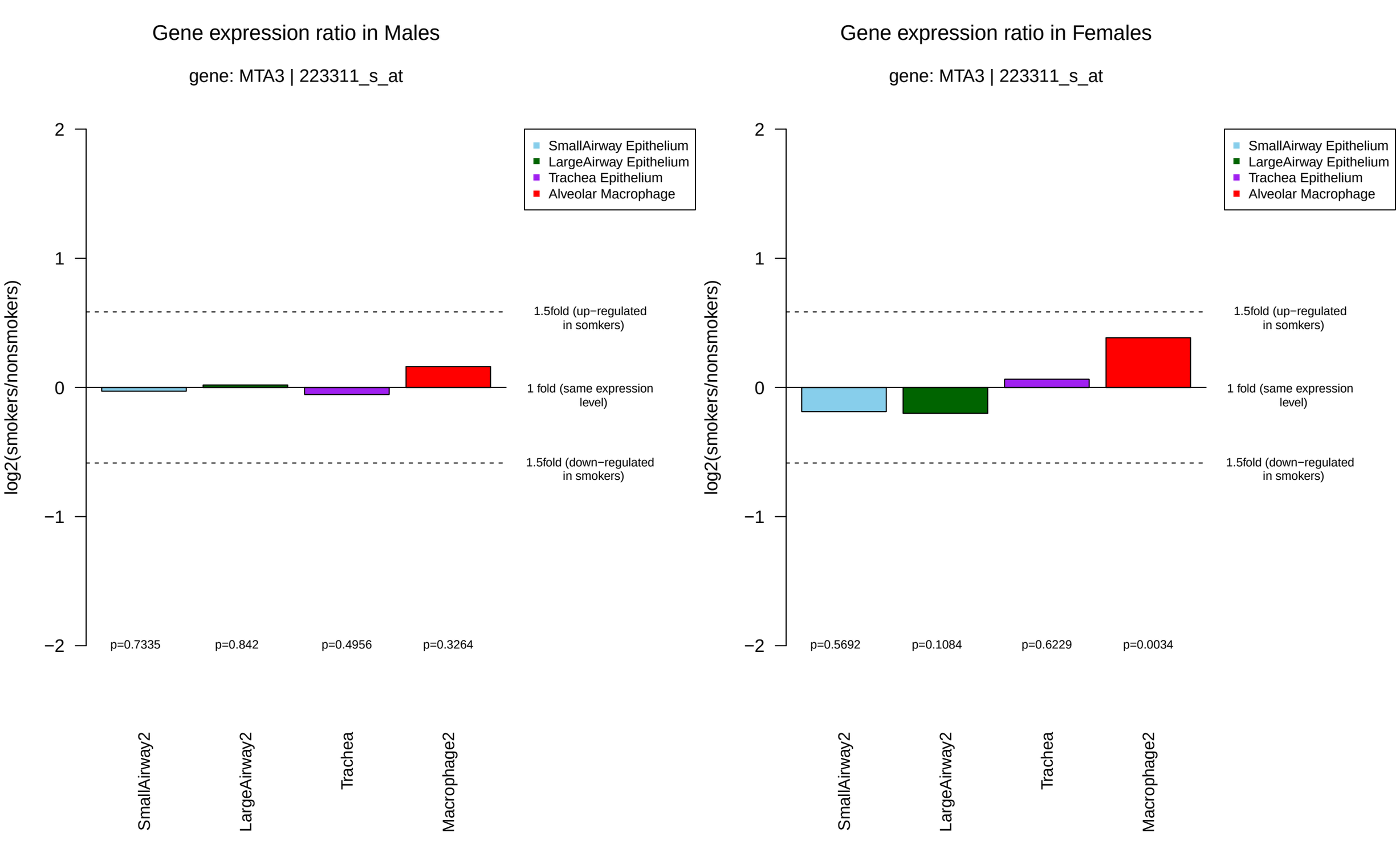
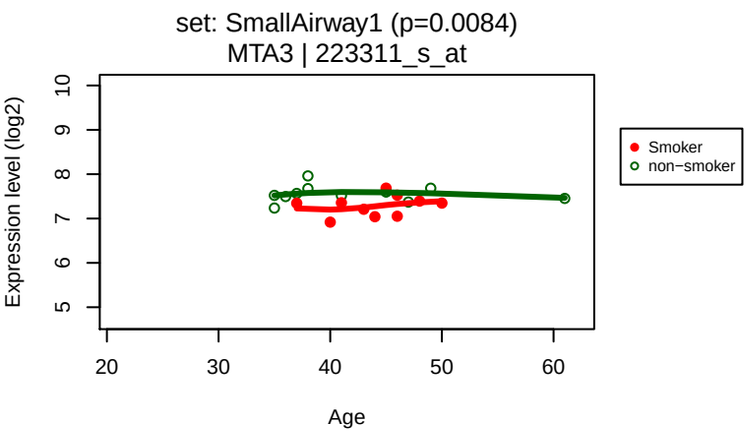
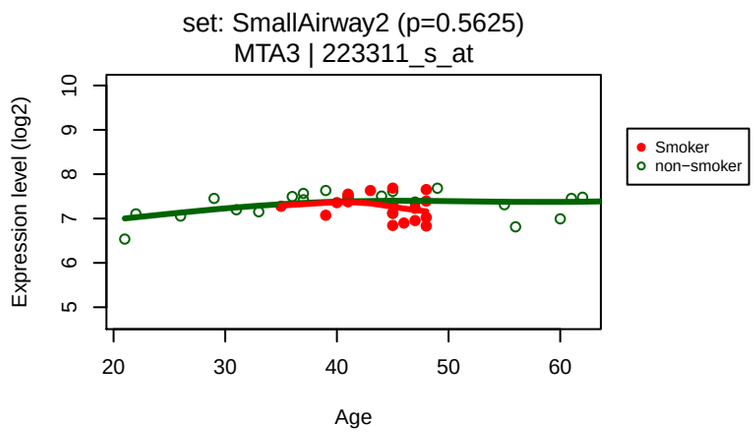
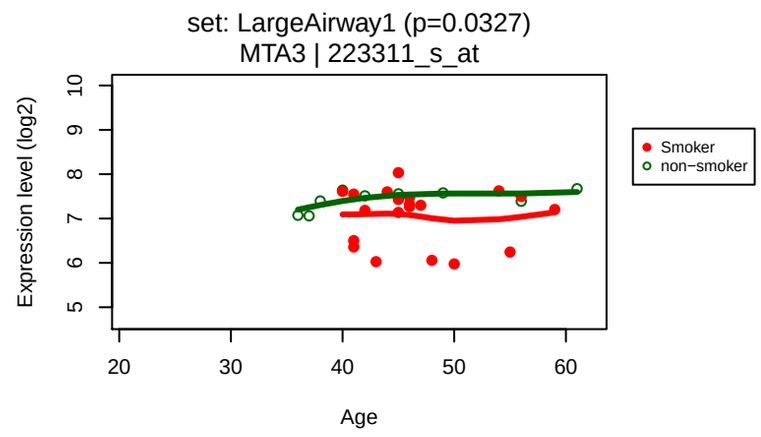

Gene expression of smokers and nonsmokers
gene: MTA3 | 223311_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0084
p=0.5625
p=0.1653
p=0.0327
p=0.5202
p=0.9209
p=0.3808
p=0.041
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

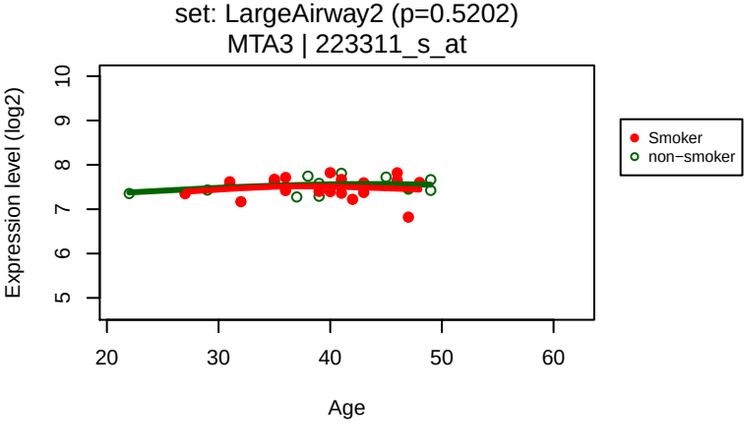

Gene expression ratio (smokers/nonsmokers)
gene: MTA3 | 223311_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0084
p=0.5625
p=0.1653
p=0.0327
p=0.5202
p=0.9209
p=0.3808
p=0.041
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

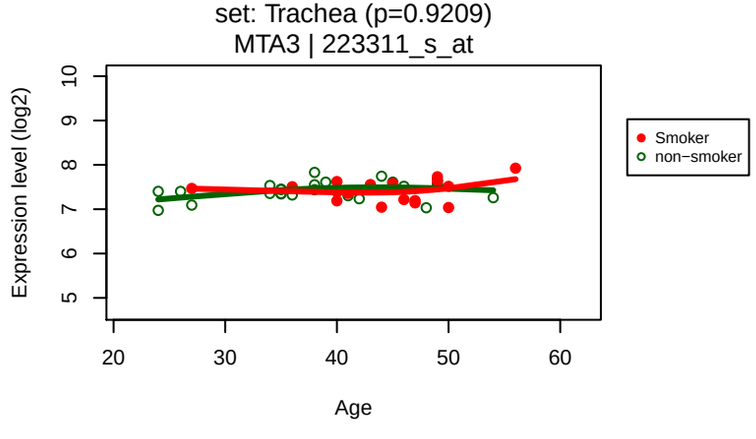

Gene expression ratio in Males
gene: MTA3 | 223311_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7335
p=0.842
p=0.4956
p=0.3264
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MTA3 | 223311_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5692
p=0.1084
p=0.6229
p=0.0034
SmallAirway2
LargeAirway2
Trachea
Macrophage2

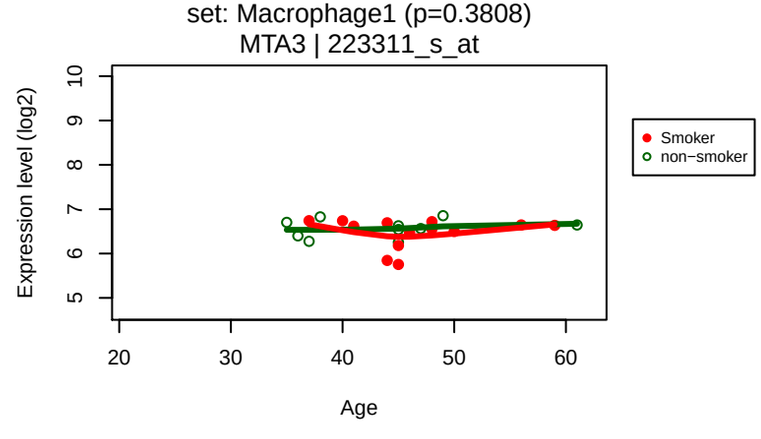

set: SmallAirway1 (p=0.0084)
MTA3 | 223311_s_at
Expression level (log2)
Age
Smoker
non−smoker

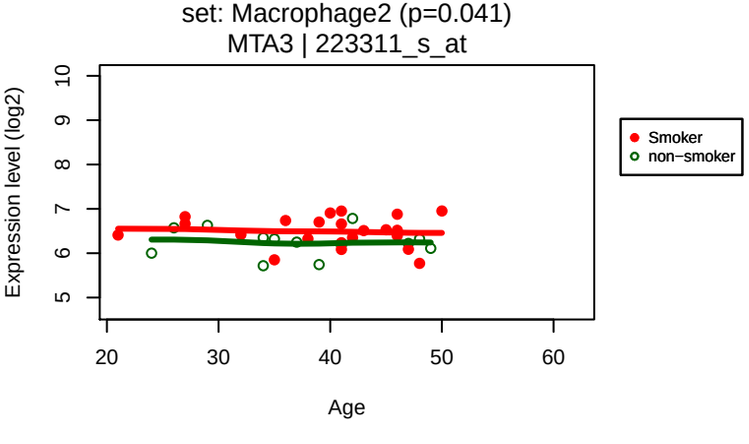

set: SmallAirway2 (p=0.5625)
MTA3 | 223311_s_at
Expression level (log2)
Age
Smoker
non−smoker

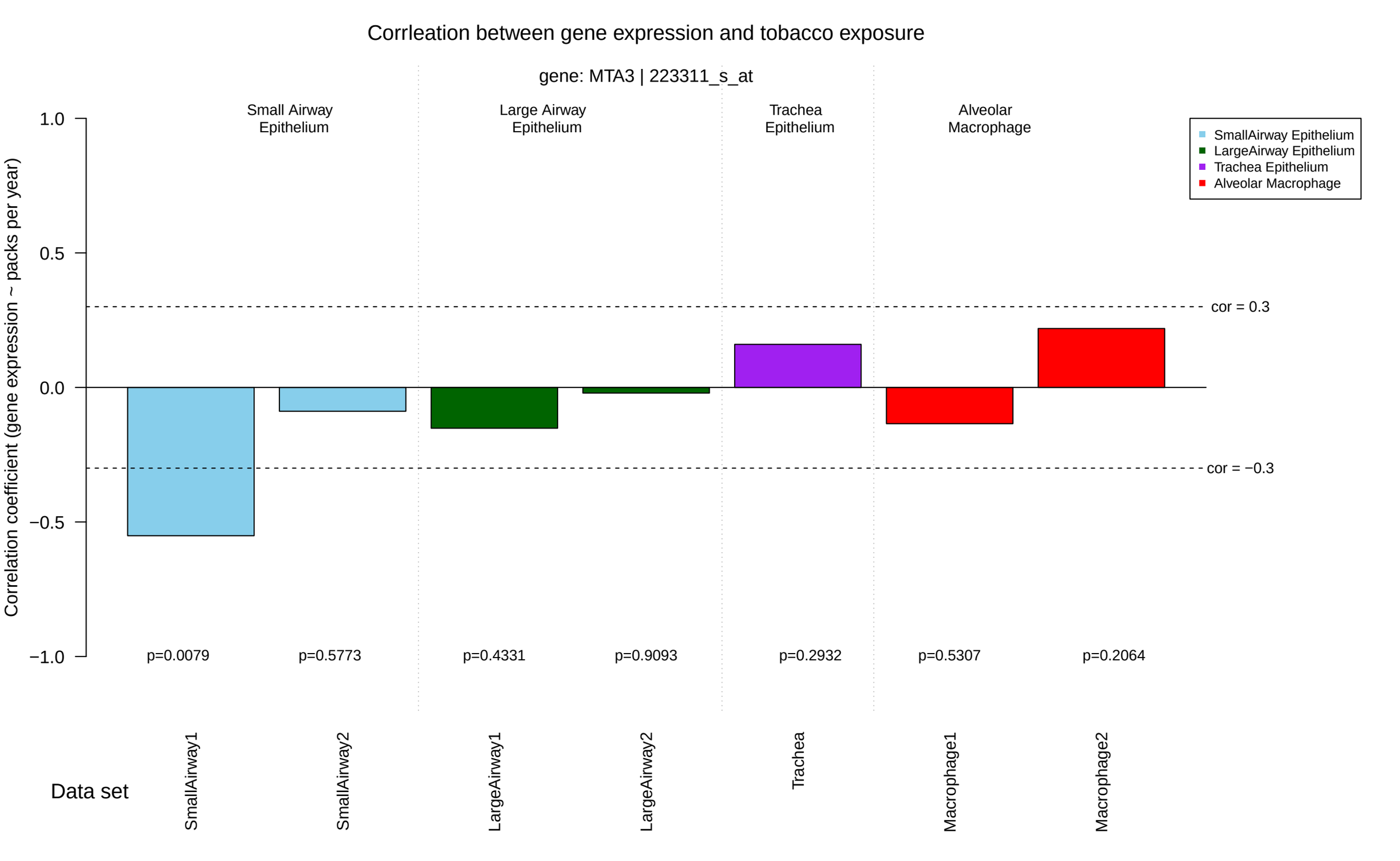

set: LargeAirway1 (p=0.0327)
MTA3 | 223311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5202)
MTA3 | 223311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9209)
MTA3 | 223311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3808)
MTA3 | 223311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.041)
MTA3 | 223311_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MTA3 | 223311_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0079
p=0.5773
p=0.4331
p=0.9093
p=0.2932
p=0.5307
p=0.2064
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2